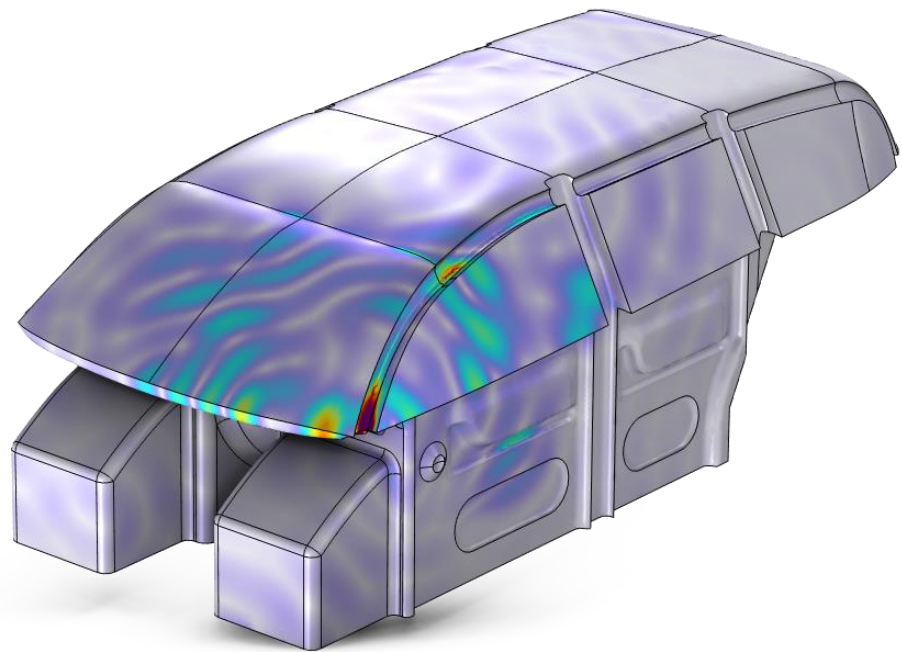




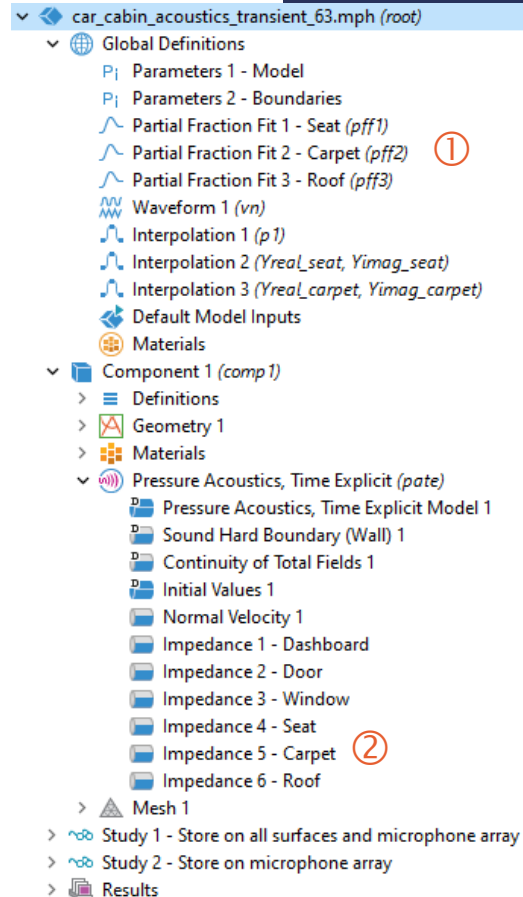
Car Cabin Acoustics — Transient Analysis

Background and Motivation

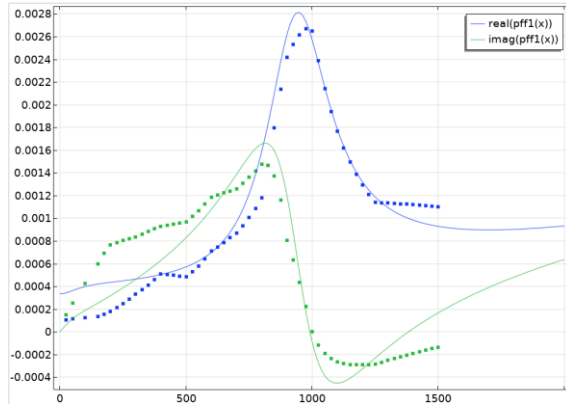
- This tutorial model shows how to model car cabin acoustics with a wave-based method in the time domain, in order to study the performance of a sound system in the low to mid frequency range. The model studies the transient response to an emitted Gaussian modulated harmonic signal solved with the *Pressure Acoustics, Time Explicit* physics interface.
- Frequency dependent impedance conditions for the seats, carpet, and roof are included using the *General local reacting (rational approximation)* impedance option. The input data is fitted using the *Partial Fraction Fit*. The model is solved using the *accelerated solver formulation* that can be run on a GPU.
- The model solves for 50 MDOFs (60 periods of the 1000 Hz signal). On a modern GPU this is done in 3h 20min.

Model Setup

- Import and fit of frequency dependent admittance for seat, carpet, and roof materials. ①
 - Fit performed with the Partial Fraction Fit function
 - Tool to convert data from frequency to time domain in a useful format
- Rational approximation of frequency dependent surface data used. ②

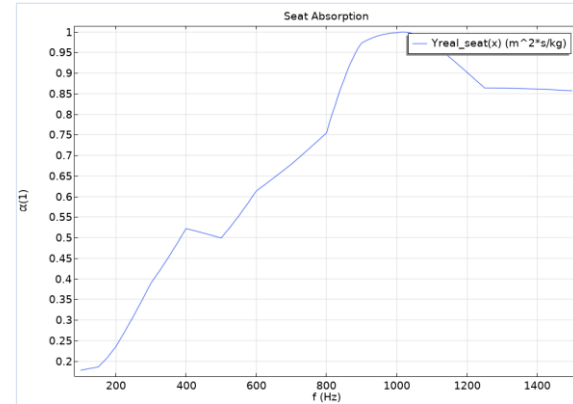


Seat Properties



Seat Admittance Data

- Measured seat admittance data (points)
- Fitted rational expansion (full lines)



Seat Absorption Coefficient

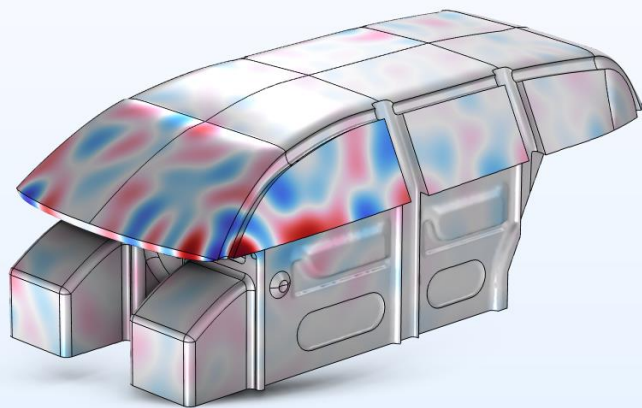
- The corresponding normal incidence absorption coefficient of the seats

Results

Time=0.02 s

Surface: Total acoustic pressure (Pa)

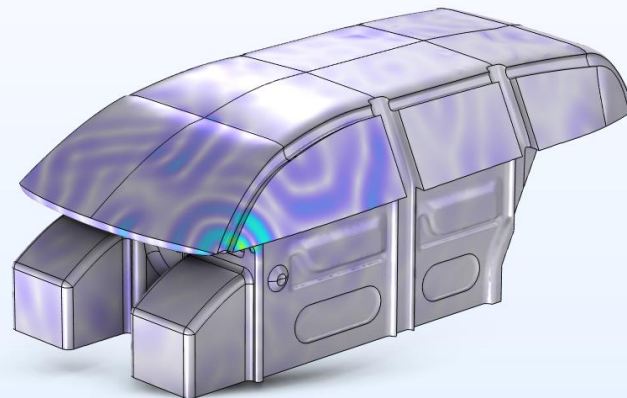
Pa



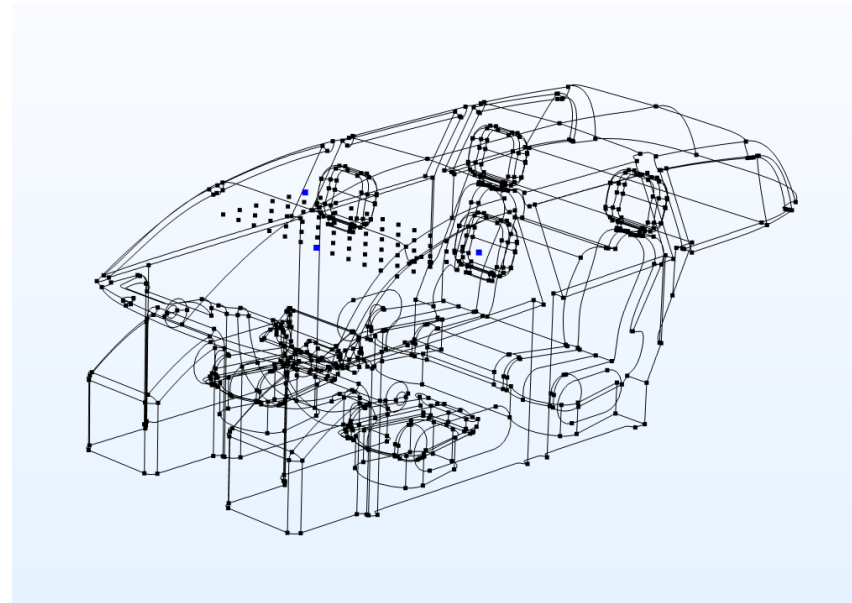
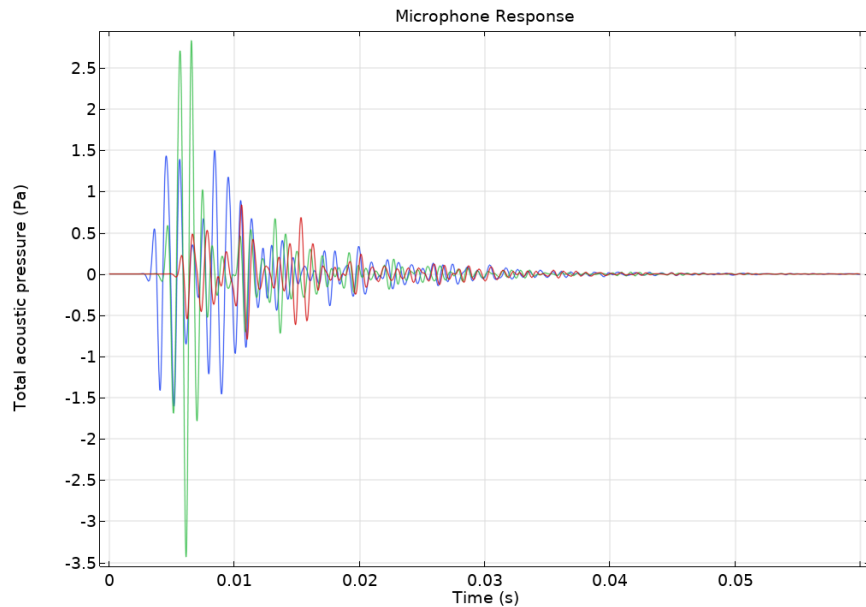
Time=0.02 s

Surface: Acoustic velocity amplitude (m/s)

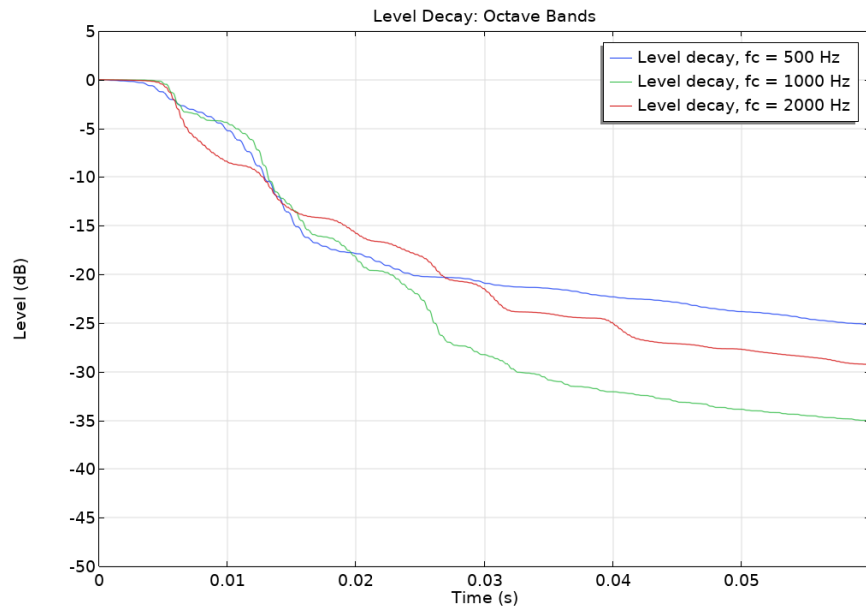
m/s
 $\times 10^{-3}$



Microphone Response (3 points on the microphone array)



Acoustic Metrics (1 microphone position)



Band center frequency (Hz)	Center time: t_s (ms)	EDT (ms)	T20 (s)
500	8.3	54.9	0.145
1000	8.6	56.6	0.046
2000	7.6	54.3	0.099